AMCG-3: UNIFIED ANTI-COMPACTION DIGEST & 3-GEAR MULTIMODAL VERIFICATION ENGINE

Eliminating Auto-Compaction Amnesia, Cascading Death Spirals, Hook Gaming & Unverified Agentic Outputs Across Multi-Agent Pipelines

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LAYER 1 • INGESTION

Universal Multi-Agent Adapter

- Normalizes lifecycle hooks across **Antigravity**, **Claude Code**, **Gemini CLI**, **Cursor** & **Codex Proxy**.
- Reconciles Undo/Rewind log branches between rewritten `transcript_full.jsonl` and append-only compact logs.
- Prevents `truncated_fields: [tool_calls]` blindness on parallel batches.

LAYER 2 • MEMORY CONTINUITY

UTF-8 Byte-Bounded Digest Synthesizer

- Distills 10.6 MB raw JSONL into ≤56.4 KB `post_compaction_digest.md` (≤345 lines).
- **Section 1:** 100% Verbatim Chronological User Prompts (Step 0 → Latest).
- **Section 2:** Canonical Turn-by-Turn Tool Calls, Edits, Errors & Replies.
- 3-byte Bangla/Unicode line bounding guarantees ≤22 KB per 120-line chunk.

LAYER 3 • ADVERSARIAL FIREWALL

Cryptographic Anti-Bypass Hook Lock

- Hard-blocks `ByteOffset > 0` overflow exploits (caught in `25f507a4`).
- Blocks 5-line micro-read probing; enforces **100% line coverage**.
- Enforces strict Conversation-UUID path isolation (blocks cross-chat spoofing).
- Header-only error detection prevents digest-body error text collisions.

LAYER 4 • ADAPTIVE ROUTING

3-Gear Risk & Cost Controller

- **Gear 1 (Fast):** Digest + Binary Crash Guard + Dead-Link Check.
- **Gear 2 (Standard):** + Automated Playwright Dual-Viewport UI & Code Execution Gate.
- **Gear 3 (Critical):** + Full 4-Tier Strict Liability & 4-File Media Multimodal QC.

-98.1% to -99.6%	0.0%	0 Loops	6 / 6 Blocked	11 Trajectories
Raw JSONL Bloat Reduction (10.6 MB → 49.8 KB)	46 KB Tool Byte-Truncation Loss (Max Chunk 22.0 KB)	Cascading Re-Compaction Death Spirals Eliminated	Adversarial Hook-Gaming Exploits Neutralized	9,165 Production Steps Forensically Audited

MODULE 1: Media & Lip-Sync Multimodal Gate

GEAR 3 • PyAV + OpenCV + Faster-Whisper

- **Signal Physics:** Hard-clamped frozen-frame (≤1.8s) and dead-silence (≤1.35s) detectors plus 40ms RMS dB cut-boundary speech clipping checks.
- **AST/Code Anti-Shortcut Scan:** Rejects FFmpeg `reverse` boomerang padding, an`ullsrc` (≥0.65s) gaps, and sine-wave camera wobble cheats.
- **4-File Multimodal Proof Bundle:** Generates 1s contact sheet, 0.5s close-up mouth/energy grid, 360p preview MP4, and master MP3 audio.

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Receipt Lock: mtime + byte-size verified across Worker → Supervisor → Chief Auditor → SI
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MODULE 2: Web & UI Visual Multimodal Gate

GEAR 2 • Headless Chromium + DOM Geometry

- **Dual-Viewport Rendering:** Automatically renders local servers or HTML builds at Desktop (`1440×900`) and Mobile (`390×844`).
- **Zero Runtime Defect Gate:** Captures uncaught `pageerror`, `console.error`, failed network assets, and broken images (`naturalWidth == 0`).
- **Layout Geometry Check:** Detects horizontal viewport overflow (`scrollWidth > innerWidth`) and produces <150 KB JPEG visual proof sheets.

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Eliminates blind "UI Complete" claims without visual & DOM console proof
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MODULE 3: Crash Firewall & Hermes Lifelong Memory

GEAR 1–3 • PyMuPDF + PIL + Self-Evolving Ledger

- **Trajectory Poisoning Guard:** Blocks `read_file` on password-encrypted PDFs, fake-extension HTML/0-byte media, and >450 KB raw images before API crash.
- **Post-Edit Practical Run Gate:** Blocks Stop hook whenever source code was edited without subsequent practical command/test verification.
- **Hermes Memory Engine:** Harvests solved bugs into `hermes_ledger.json` with regex pre-tool firewalls so historical mistakes never recur.

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Permanent Cross-Session Immunity Across All Connected Agentic Tools
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